

Introduction to RNA Sequencing

Part 3: Differential Expression

Adapted from RNAbio.org

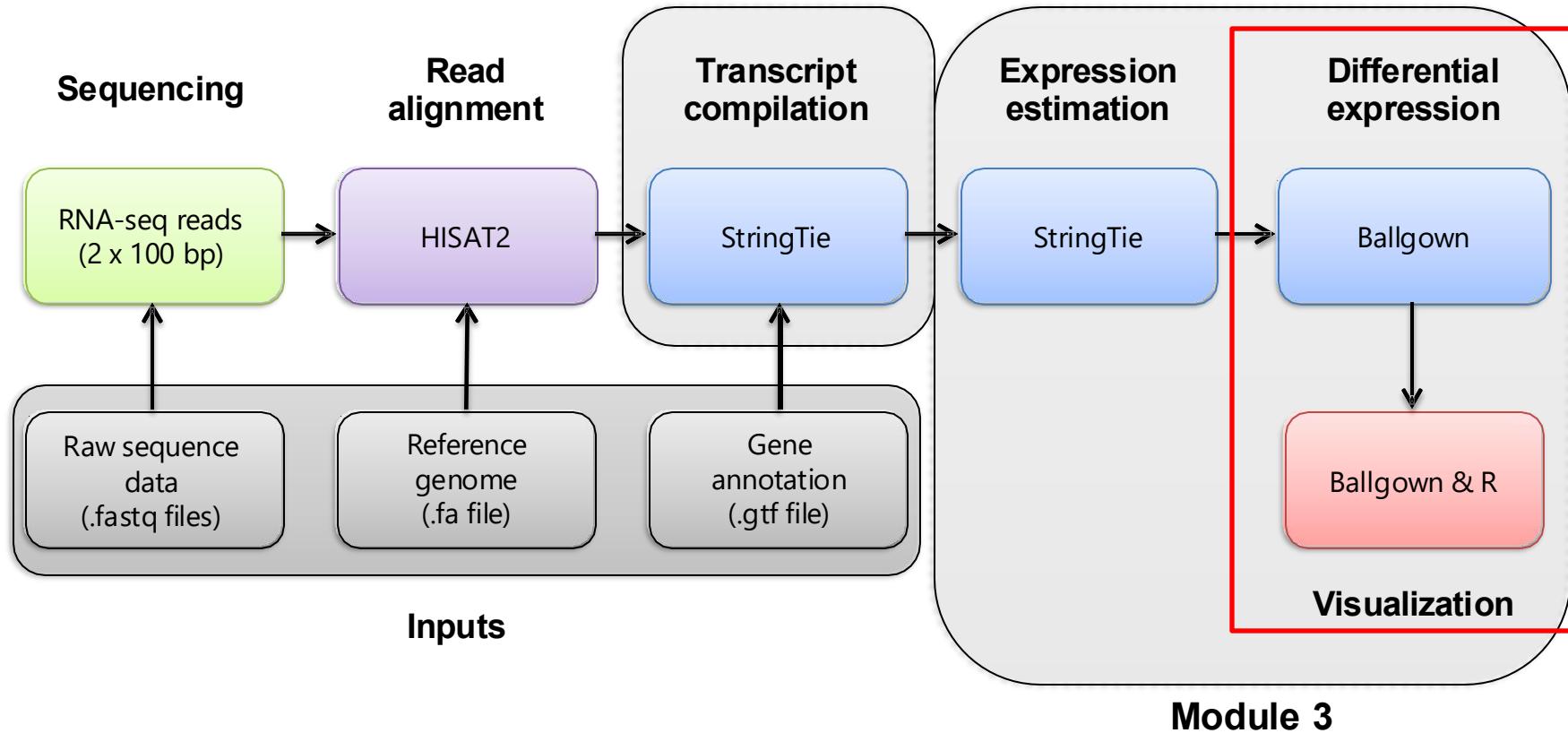
Material created by:

Arpad Danos, Felicia Gomez, Obi Griffith, Malachi Griffith,
My Hoang, Mariam Khanfar, Chris Miller, Kartik Singhal

RNAseq modules

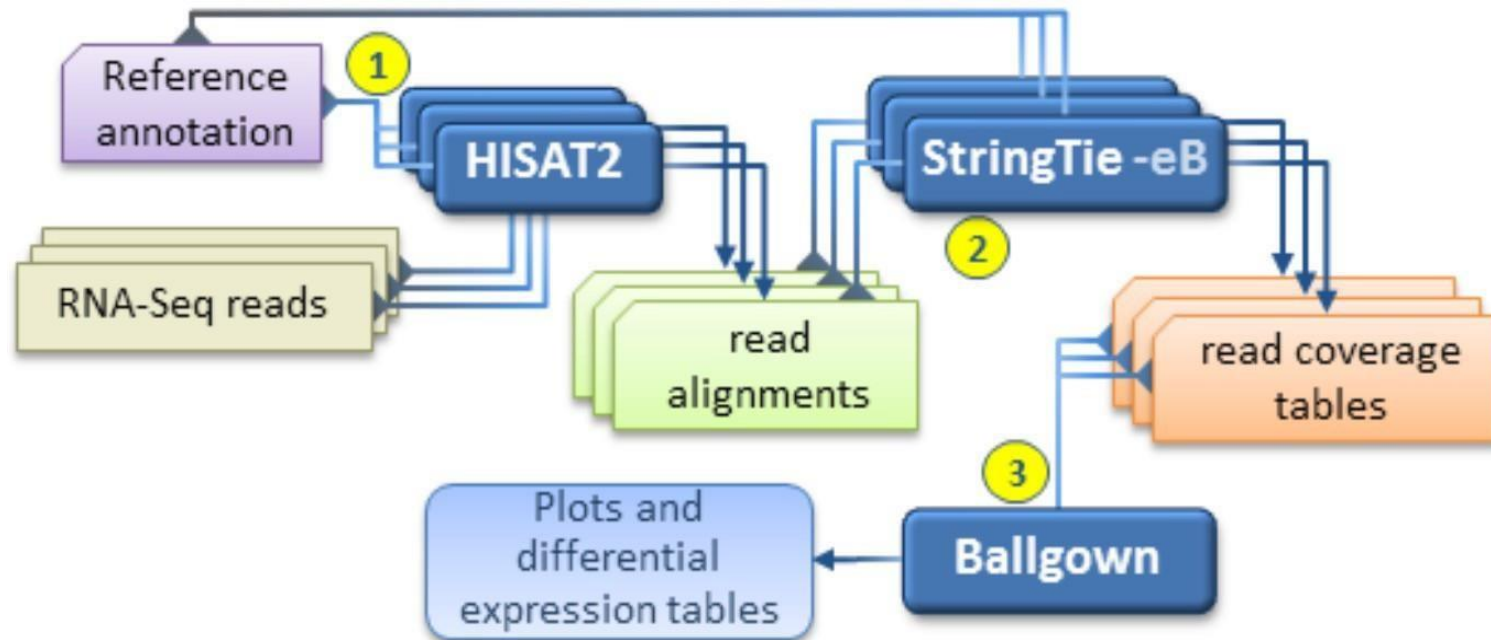
- RNA study design and spliced alignment
- Gene/transcript abundance estimation
- **Differential expression methods**

HISAT2/StringTie/Ballgown RNA-seq Pipeline



Last week: Expression estimation (StringTie, htseq count)
This week: Differential expression (Ballgown, edgeR)

Stringtie_Expression Estimate



This is the workflow we used in last week's exercise:
StringTie -G and -e

Expression estimation mode ("Reference Only")

RNA Input Data

- Universal Human Reference (UHR) and Human Brain Reference (HBR)
- In addition, a spike-in control was used (ERCC ExFold RNA Spike-In Control Mixes) to each sample.
 - UHR + ERCC Spike-In Mix1, Replicate 1
 - UHR + ERCC Spike-In Mix1, Replicate 2
 - UHR + ERCC Spike-In Mix1, Replicate 3
 - HBR + ERCC Spike-In Mix2, Replicate 1
 - HBR + ERCC Spike-In Mix2, Replicate 2
 - HBR + ERCC Spike-In Mix2, Replicate 3

Purpose of ERCC in RNAs and RNA-Seq Analysis

Normalization: control for variations in RNA input, reverse transcription efficiency, PCR amplification biases, and sequencing depth

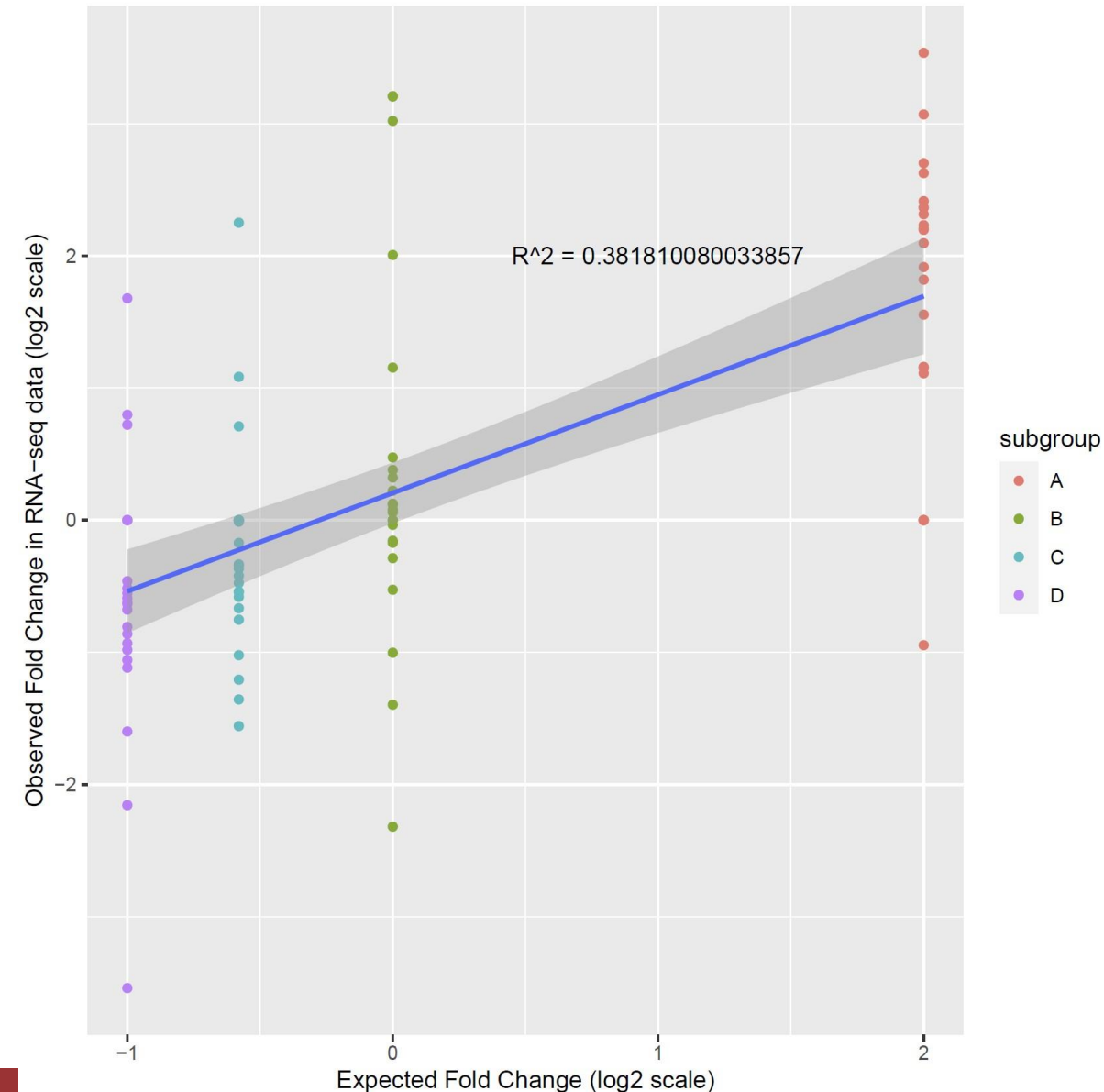
Validation: provide a way to assess the accuracy of RNA-Seq measurements by comparing the observed fold changes to the known, expected fold changes

Quality Control: used to check the overall performance of the RNA-Seq workflow, from library preparation to sequencing and data analysis

Table 1 Transcript molar ratios in ERCC Spike-In Mixes

Subgroup	Mix 1:Mix 2 [†]
A	4.00
B	1.00
C	0.67
D	0.50

[†] Applies only to Spike-In Mix 1 and Mix 2 with same manufacturing lot number.



Reminders:

- Normalized counts account for sequencing depth and gene length biases
 - RPKM $\sim$ single-end sequencing, FPKM $\sim$ paired-end sequencing
 - CPM = counts per million, can compare single genes across samples
 - The sum of all TPMs in each sample is the same. Easier to compare across samples!
- Abundance estimation tool that calculates normalized count (FPKM, TPM): StringTie
- Abundance estimation tool that calculates raw count: HTseq

Differential Expression

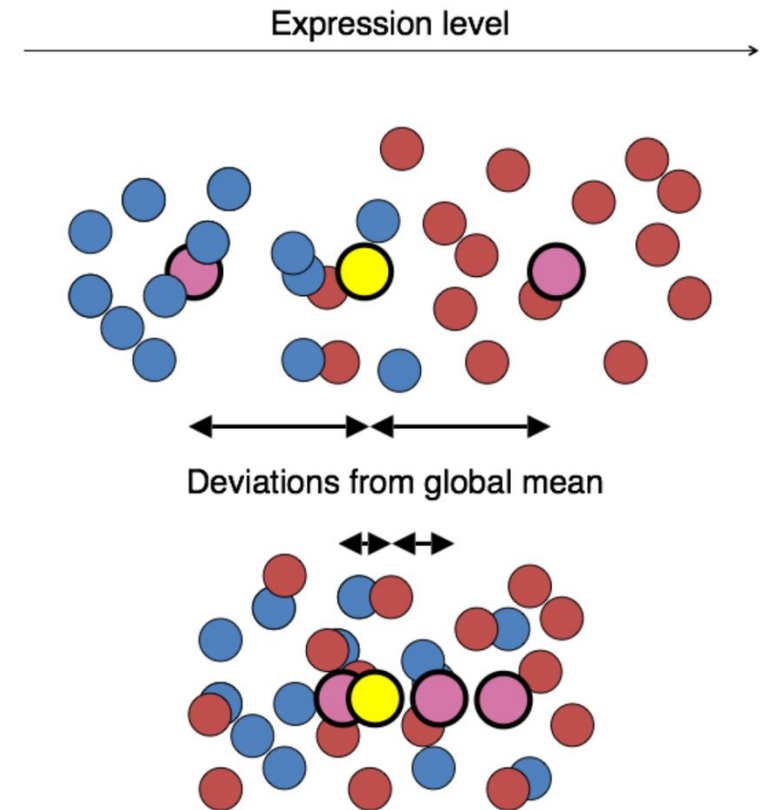
- Tying gene expression back to genotype/phenotype

- What genes/transcripts are being expressed at higher/lower levels in different groups of samples?

- Are these differences 'significant', accounting for variance/noise?

- Examples (used in course):

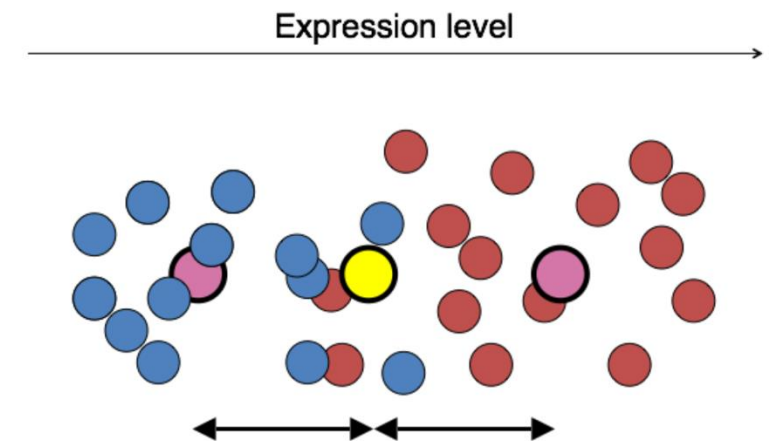
- UHR cells vs HBR brain
 - Tumor vs Normal tissue
 - Wild-type vs gene KO cells



Differential Expression with Ballgown

Parametric F-test comparing nested linear models

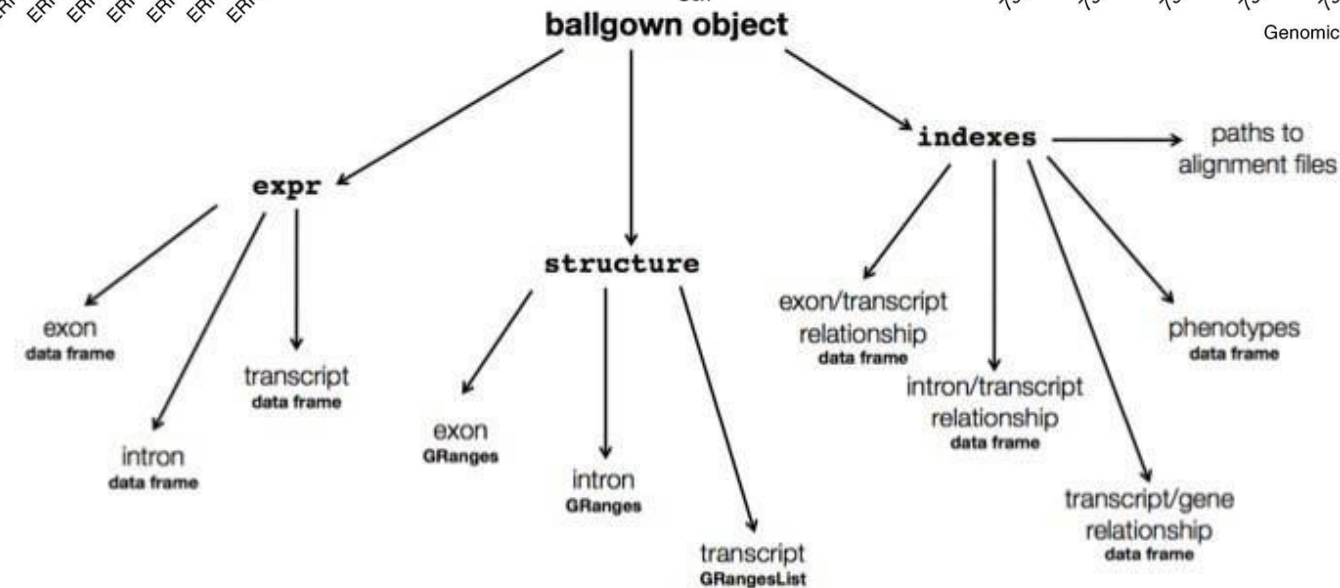
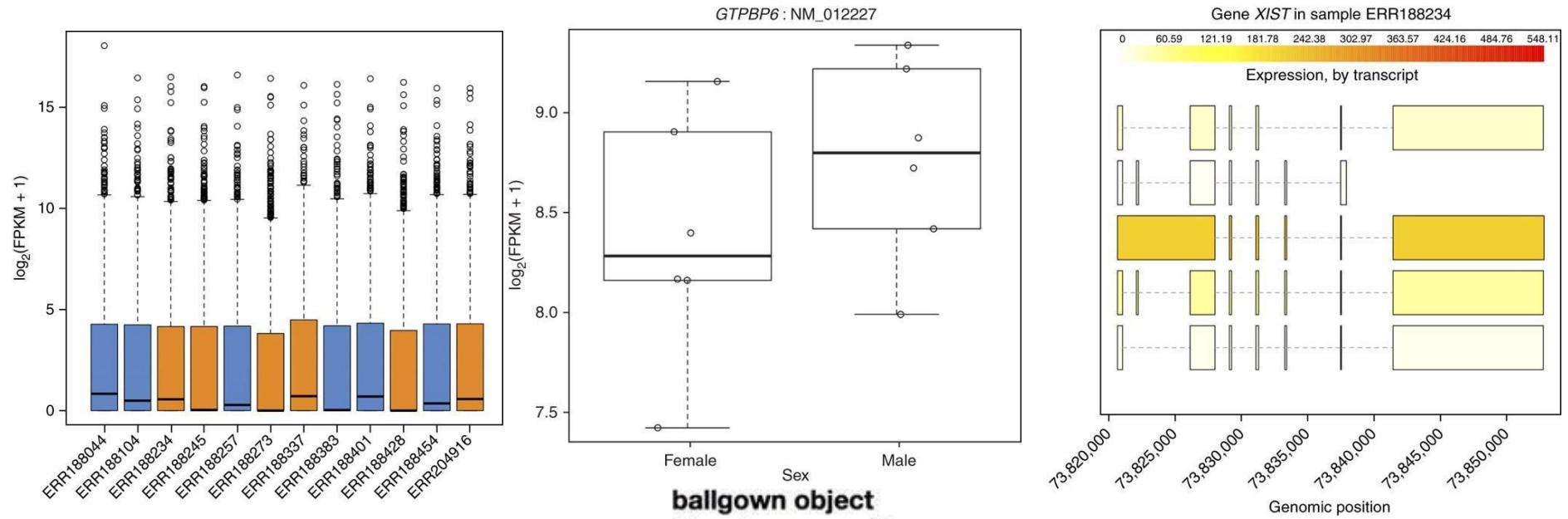
- Two models are fit to each feature, using expression as the outcome
 - one including the covariate of interest (e.g., case/control status or time) and one not including that covariate.
- An F statistic and p-value are calculated using the fits of the two models.
 - A significant p-value means the model including the covariate of interest fits significantly better than the model without that covariate, indicating differential expression.
- Adjust for multiple testing by reporting q-values:
 - $q < 0.05$ the false discovery rate is controlled at ~5%.



Multiple testing correction

- As more attributes are compared, differences due solely to chance become more likely!
- Well known from array studies
 - 10,000s genes/transcripts
 - 100,000s exons
- With RNA-seq, more of a problem than ever
 - All the complexity of the transcriptome gives huge numbers of potential features
 - Genes, transcripts, exons, junctions, retained introns, microRNAs, lncRNAs, etc
- Bioconductor multtest
 - <http://www.bioconductor.org/packages/release/bioc/html/multtest.html>

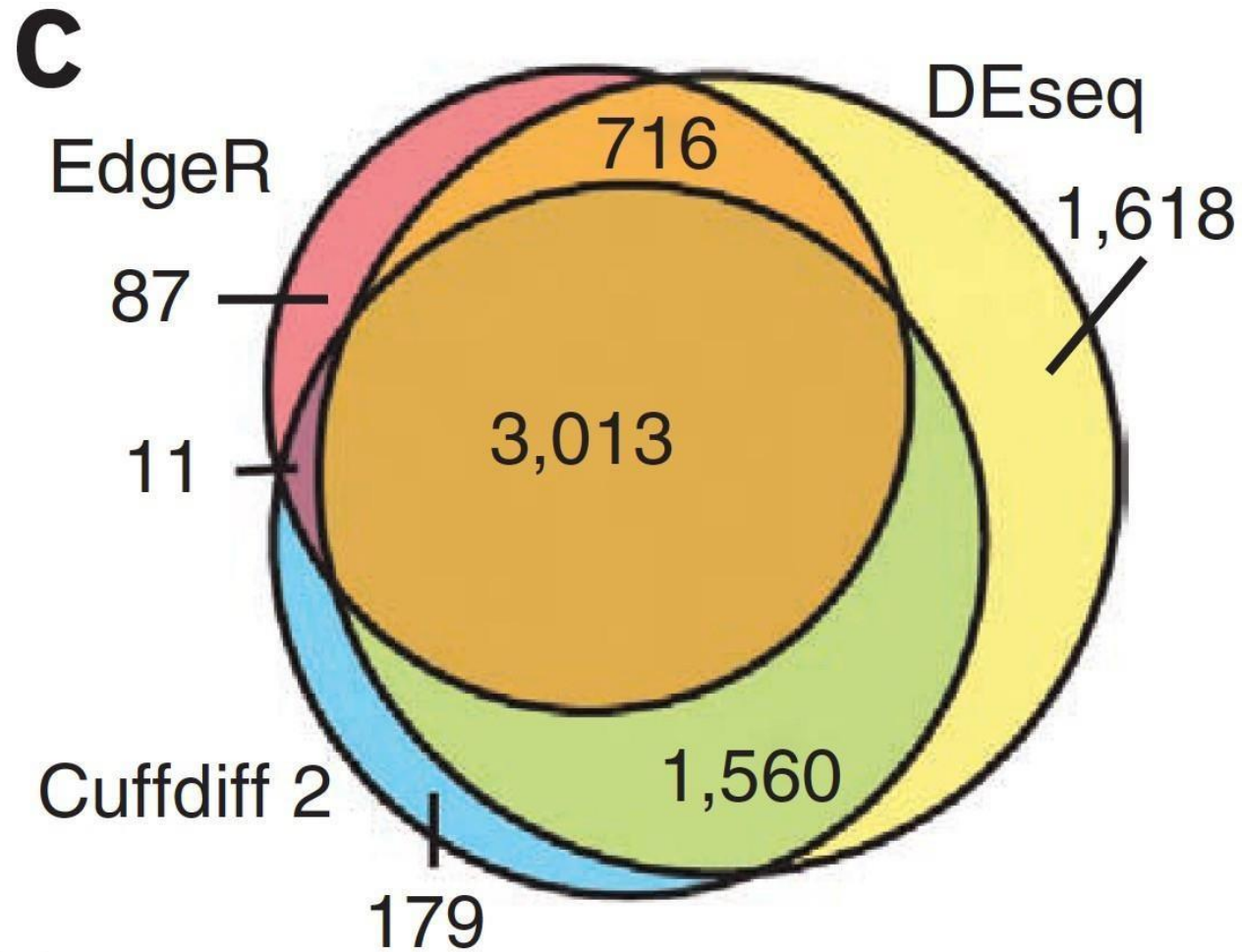
Ballgown is one tool for DE



Alternative differential expression methods

- Raw count approaches
 - DESeq2 - <http://www-huber.embl.de/users/anders/DESeq/>
 - edgeR - <http://www.bioconductor.org/packages/release/bioc/html/edgeR.html>
 - Others...

Multiple approaches advisable



Power depends on study design

- Hansen et al. “Sequencing Technology Does Not Eliminate Biological Variability.” Nature Biotechnology 29, no. 7 (2011): 572–573.
- Power analysis for RNA-seq experiments
 - https://rafalwoycicki.github.io/power_calculator/power_calculator.html
- RNA-seq need for biological replicates
 - <http://www.biostars.org/p/1161/>
- RNA-seq study design
 - <http://www.biostars.org/p/68885/>

Assignment: Ballgown DE Analysis

- Run R and load required libraries

```
ubuntu@c6c4b48da477:~/workspace/rnaseq/de/ballgown/ref_only$ pwd
/home/ubuntu/workspace/rnaseq/de/ballgown/ref_only
ubuntu@c6c4b48da477:~/workspace/rnaseq/de/ballgown/ref_only$ ls
ubuntu@c6c4b48da477:~/workspace/rnaseq/de/ballgown/ref_only$ R

R version 4.0.0 (2020-04-24) -- "Arbor Day"
Copyright (C) 2020 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
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Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> library(ballgown)

Attaching package: 'ballgown'

The following object is masked from 'package:base':

  structure

> library(genefilter)
> library(dplyr)

Attaching package: 'dplyr'

The following objects are masked from 'package:ballgown':

  contains, expr, last

The following objects are masked from 'package:stats':

  filter, lag

The following objects are masked from 'package:base':

  intersect, setdiff, setequal, union

> library(devtools)
Loading required package: usethis
> |
```

Create phenotype data needed for ballgown analysis

```
> getwd()
[1] "/workspace/rnaseq/de/ballgown/ref_only"
> results="/home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/"
> results
[1] "/home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/"
> path=paste(results,ids,sep="")
> pheno_data=data.frame(ids,type,path)
> path
[1] "/home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/UHR_Rep1"
[2] "/home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/UHR_Rep2"
[3] "/home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/UHR_Rep3"
[4] "/home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/HBR_Rep1"
[5] "/home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/HBR_Rep2"
[6] "/home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/HBR_Rep3"
> pheno_data
  ids type
1 UHR_Rep1 UHR
2 UHR_Rep2 UHR
3 UHR_Rep3 UHR
4 HBR_Rep1 HBR
5 HBR_Rep2 HBR
6 HBR_Rep3 HBR

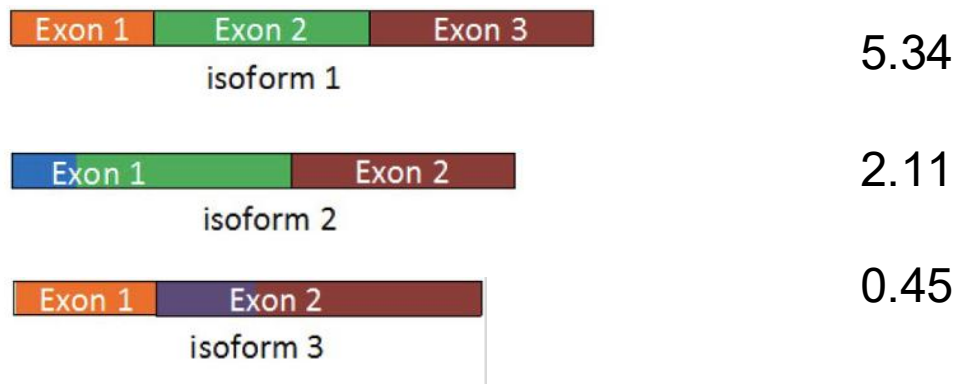
                                path
1 /home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/UHR_Rep1
2 /home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/UHR_Rep2
3 /home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/UHR_Rep3
4 /home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/HBR_Rep1
5 /home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/HBR_Rep2
6 /home/ubuntu/workspace/rnaseq/expression/stringtie/ref_only/HBR_Rep3
```

Outputs of StringTie

GTF file with:

Transcript structure

Transcript Quantification



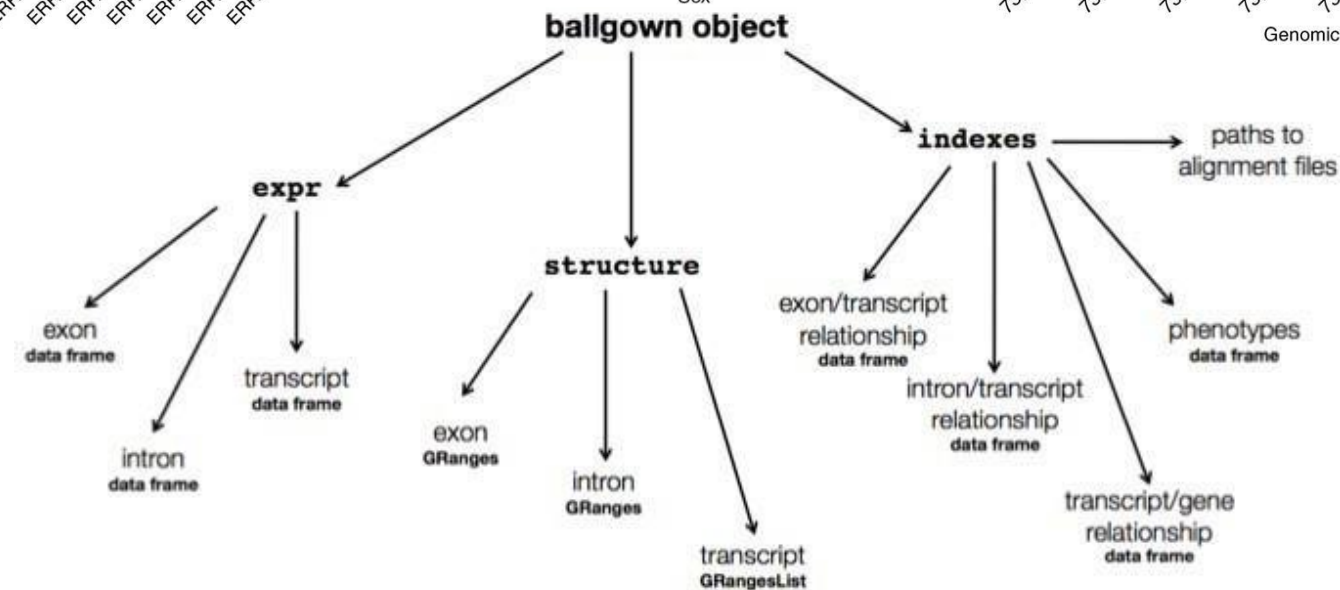
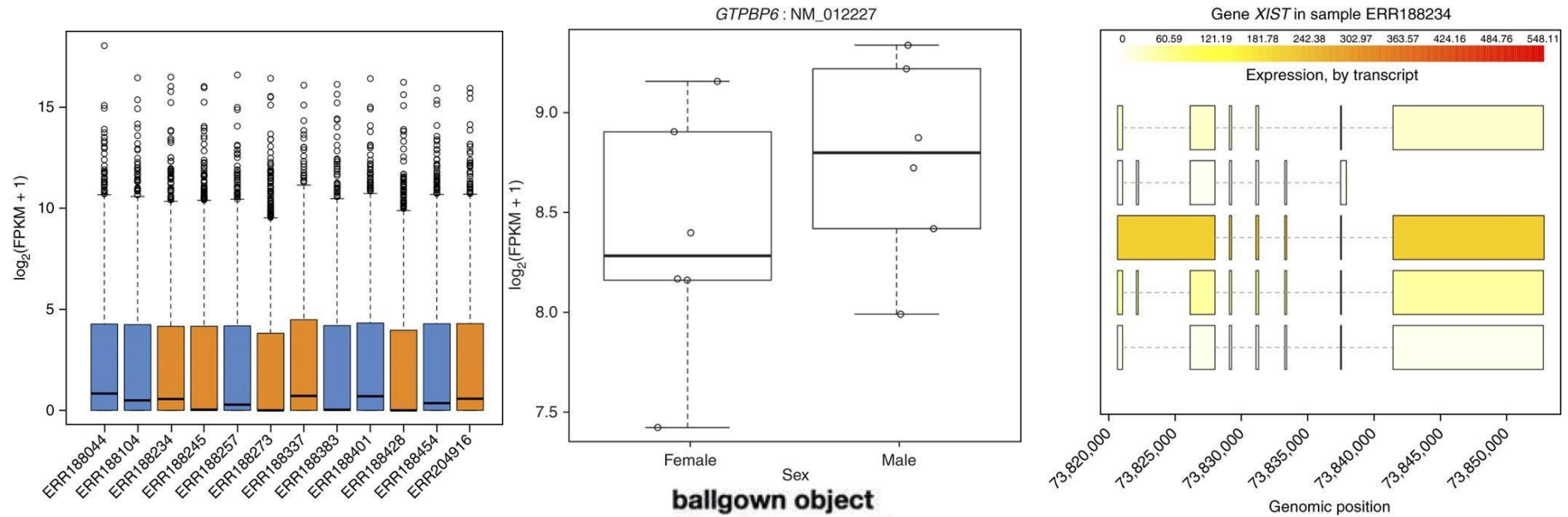
```
chr1 StringTie transcript 114704469 114716894 1000 - . gene_id "ENSG00000213281";  
transcript_id "ENST00000369535"; ref_gene_name "NRAS"; cov "111.583565"; FPKM "17.794451"; TPM "34.907570";
```

Gene expression file (summing the transcripts):

Gene ID	Gene Name	Reference	Strand	Start	End	Coverage	FPKM	TPM
ENSG00000213281	NRAS	chr1	-	114704469	114716894	81.372406	21.631533	42.434814

Pertea et al. Nature Biotechnology, 2015

Ballgown for Visualization with R



Create the ballgown data structure

```
> bg = ballgown(samples = as.vector(pheno_data$path), pData = pheno_data)
Mon Dec 11 03:12:48 2023
Mon Dec 11 03:12:48 2023: Reading linking tables
Mon Dec 11 03:12:48 2023: Reading intron data files
Mon Dec 11 03:12:49 2023: Merging intron data
Mon Dec 11 03:12:49 2023: Reading exon data files
Mon Dec 11 03:12:51 2023: Merging exon data
Mon Dec 11 03:12:51 2023: Reading transcript data files
Mon Dec 11 03:12:52 2023: Merging transcript data
Wrapping up the results
```

```
> bg
ballgown instance with 4564 transcripts and 6 samples
```

Verify your object

Attributes

- Extract all transcript-level expression data from the bg object. Then extract unique gene and unique transcript IDs.

```
> head(bg_table)
  t_id chr strand    start      end      t_name num_exons length
1    1  22     - 10736171 10736283 ENST000000615943         1    113
2    2  22     - 10939388 10961338 ENST000000635667         9    749
3    3  22     - 11065974 11067346 ENST000000623473         2     54
4    4  22     + 11066501 11068089 ENST000000624155         2    120
5    5  22     + 11124337 11125705 ENST000000422332         2   1241
6    6  22     - 11249809 11249959 ENST000000612732         1    151

  gene_id gene_name cov.UHR_Rep1 FPKM.UHR_Rep1 cov.UHR_Rep2
1 ENSG00000277248      U2              0              0              0
2 ENSG00000283047   FRG1FP              0              0              0
3 ENSG00000280363 CU104787.1              0              0              0
4 ENSG00000279973    BAGE5              0              0              0
5 ENSG00000226444  ACTR3BP6              0              0              0
6 ENSG00000276871  5_8S_rRNA              0              0              0

  FPKM.UHR_Rep2 cov.UHR_Rep3 FPKM.UHR_Rep3 cov.HBR_Rep1 FPKM.HBR_Rep1
1              0              0              0              0              0
2              0              0              0              0              0
3              0              0              0              0              0
4              0              0              0              0              0
5              0              0              0              0              0
6              0              0              0              0              0

  cov.HBR_Rep2 FPKM.HBR_Rep2 cov.HBR_Rep3 FPKM.HBR_Rep3
1              0              0              0              0
2              0              0              0              0
3              0              0              0              0
4              0              0              0              0
5              0              0              0              0
6              0              0              0              0
```

```
> head(bg_gene_names)
  gene_id gene_name
1 ENSG00000277248      U2
2 ENSG00000283047   FRG1FP
3 ENSG00000280363 CU104787.1
4 ENSG00000279973    BAGE5
5 ENSG00000226444  ACTR3BP6
6 ENSG00000276871  5_8S_rRNA
```

```
> head(bg_transcript_names)
  t_id      t_name
1    1 ENST000000615943
2    2 ENST000000635667
3    3 ENST000000623473
4    4 ENST000000624155
5    5 ENST000000422332
6    6 ENST000000612732
```

Perform DE analysis

```
results_transcripts = stattest(bg, feature="transcript", covariate="type",  
getFC=TRUE, meas="FPKM")
```

```
results_transcripts = merge(results_transcripts, bg_transcript_names, by.x=c("id"),  
by.y=c("t_id"))
```

```
> head(results_transcripts)
  id feature      fc      pval      qval      t_name
1  1 transcript 1.0000000      NaN      NaN ENST00000615943
2 10 transcript 1.0000000      NaN      NaN ENST00000448473
3 100 transcript 1.0000000      NaN      NaN ENST00000517943
4 1000 transcript 1.0000000      NaN      NaN ENST00000403807
5 1001 transcript 1.0000000      NaN      NaN ENST00000302273
6 1002 transcript 0.8876485 0.8829198 0.955367 ENST00000624350
> head(results_genes)
      id feature      fc      pval      qval gene_name
1 ENSG00000008735    gene 0.01383563 0.0002270410 0.003574835 MAPK8IP2
2 ENSG00000015475    gene 1.58883098 0.0054844568 0.026638790      BID
3 ENSG00000025708    gene 1.39579593 0.3992876858 0.596233639      TYMP
4 ENSG00000025770    gene 1.46572045 0.0316457273 0.103699543    NCAPH2
5 ENSG00000040608    gene 0.10280538 0.0004183902 0.005360246      RTN4R
6 ENSG00000054611    gene 1.09845192 0.1808088853 0.352059592    TBC1D22A
```

Filter low-abundance genes

`subset()`: This function subsets the `bg` object to include only those transcripts with a variance across the samples greater than 1.

This step is designed to remove transcripts that do not show much change across your conditions, under the assumption that they are not likely to be biologically interesting

`bg_filt = subset (bg, "rowVars(texpr(bg)) > 1", genomesubset=TRUE)`

where `genomesubset=TRUE` → ensures that when you subset the transcripts, the associated genomic features (like exons and introns) are also appropriately subsetted.

```
> nrow(bg_table)
[1] 4564
> nrow(bg_filt_table)
[1] 2924
> nrow(bg_gene_names)
[1] 1410
> nrow(bg_filt_gene_names)
[1] 830
> nrow(bg_transcript_names)
[1] 4564
> nrow(bg_filt_transcript_names)
[1] 2924
```

Perform DE analysis using the filtered data and identify significant genes

```
results_transcripts = statstest(bg_filt, feature="transcript", covariate="type", getFC=TRUE, meas="FPKM") <- for the filtered one...
```

```
sig_transcripts = subset(results_transcripts, results_transcripts$pval<0.05)
```

```
sig_genes = subset(results_genes, results_genes$pval<0.05)
```

```
> head(sig_transcripts)
```

	id	feature	fc	pval	qval	t_name
13	1035	transcript	32.66479	0.0006079048	0.03192359	ENST00000302097
14	1036	transcript	517.14266	0.0134847913	0.18801647	ENST00000398743
16	1038	transcript	1314.56328	0.0060651295	0.11368230	ENST00000398741
17	1039	transcript	19.19580	0.0227337966	0.24896487	ENST00000543184
18	1040	transcript	930.65593	0.0175994799	0.21084735	ENST00000405655
19	1041	transcript	60.37137	0.0115773205	0.17271472	ENST00000402697

```
> head(sig_genes)
```

	id	feature	fc	pval	qval	gene_name
1	ENSG00000008735	gene	0.01411366	0.0001930115	0.003076761	MAPK8IP2
2	ENSG00000015475	gene	1.57305207	0.0056168528	0.025899932	BID
4	ENSG00000025770	gene	1.47840611	0.0248558798	0.079347616	NCAPH2
5	ENSG00000040608	gene	0.10382267	0.0004159533	0.004810006	RTN4R
8	ENSG00000069998	gene	2.57657413	0.0068230552	0.029342673	CECR5
9	ENSG00000070010	gene	2.18390717	0.0009926846	0.007772907	UFD1L

Expected output in

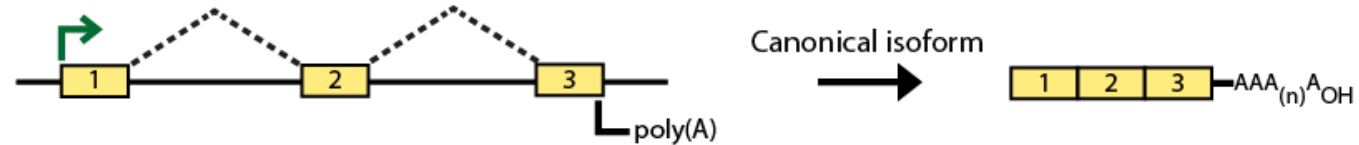
\$RNA_HOME/de/ballgown/ref_only/

```
ubuntu@c6c4b48da477:~/workspace/rnaseq/de/ballgown/ref_only$ pwd
/home/ubuntu/workspace/rnaseq/de/ballgown/ref_only
ubuntu@c6c4b48da477:~/workspace/rnaseq/de/ballgown/ref_only$ ls -alt
total 2868
-rw-rw-r-- 1 ubuntu ubuntu 26648 Dec 11 04:00 UHR_vs_HBR_gene_results_sig.tsv
drwxrwxr-x 1 ubuntu ubuntu 4096 Dec 11 04:00 .
-rw-rw-r-- 1 ubuntu ubuntu 38495 Dec 11 04:00 UHR_vs_HBR_transcript_results_sig.tsv
-rw-rw-r-- 1 ubuntu ubuntu 69678 Dec 11 04:00 UHR_vs_HBR_gene_results_filtered.tsv
-rw-rw-r-- 1 ubuntu ubuntu 249590 Dec 11 03:59 UHR_vs_HBR_transcript_results_filtered.tsv
-rw-rw-r-- 1 ubuntu ubuntu 95058 Dec 11 03:48 UHR_vs_HBR_gene_results.tsv
-rw-rw-r-- 1 ubuntu ubuntu 325014 Dec 11 03:48 UHR_vs_HBR_transcript_results.tsv
-rw-rw-r-- 1 ubuntu ubuntu 2114225 Dec 11 03:42 bg.rda
drwxrwxr-x 1 ubuntu ubuntu 4096 Dec 11 03:02 ..
```

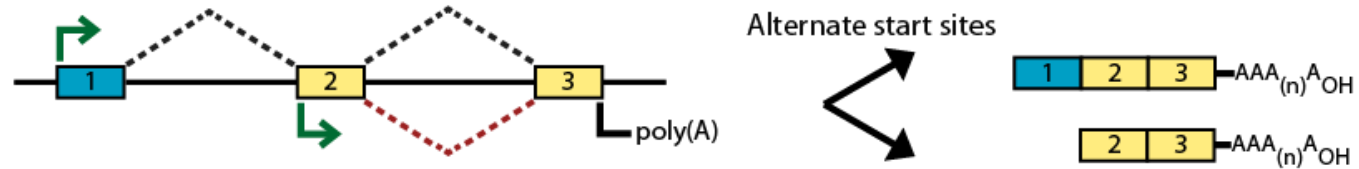
Differential transcript usage

aka alternative splicing

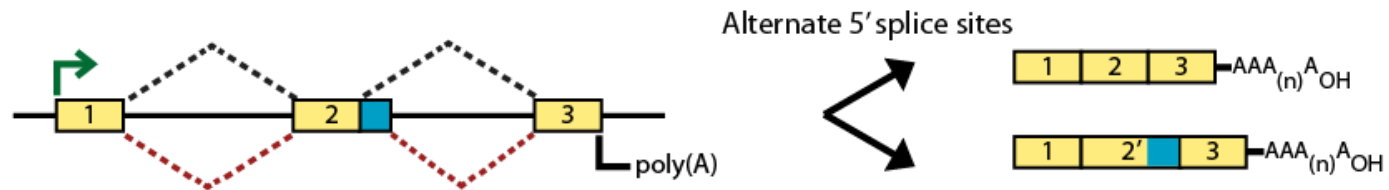
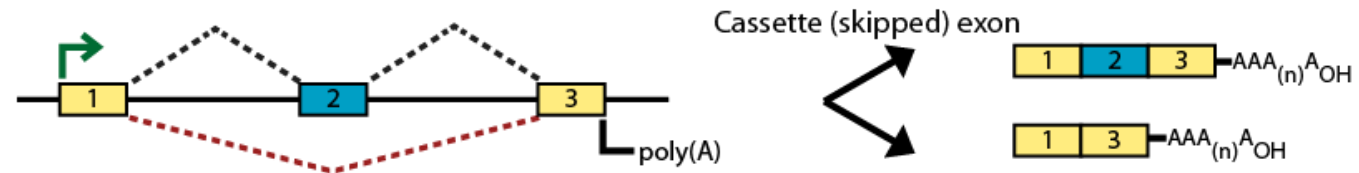
Simple transcription



Alternative transcript initiation

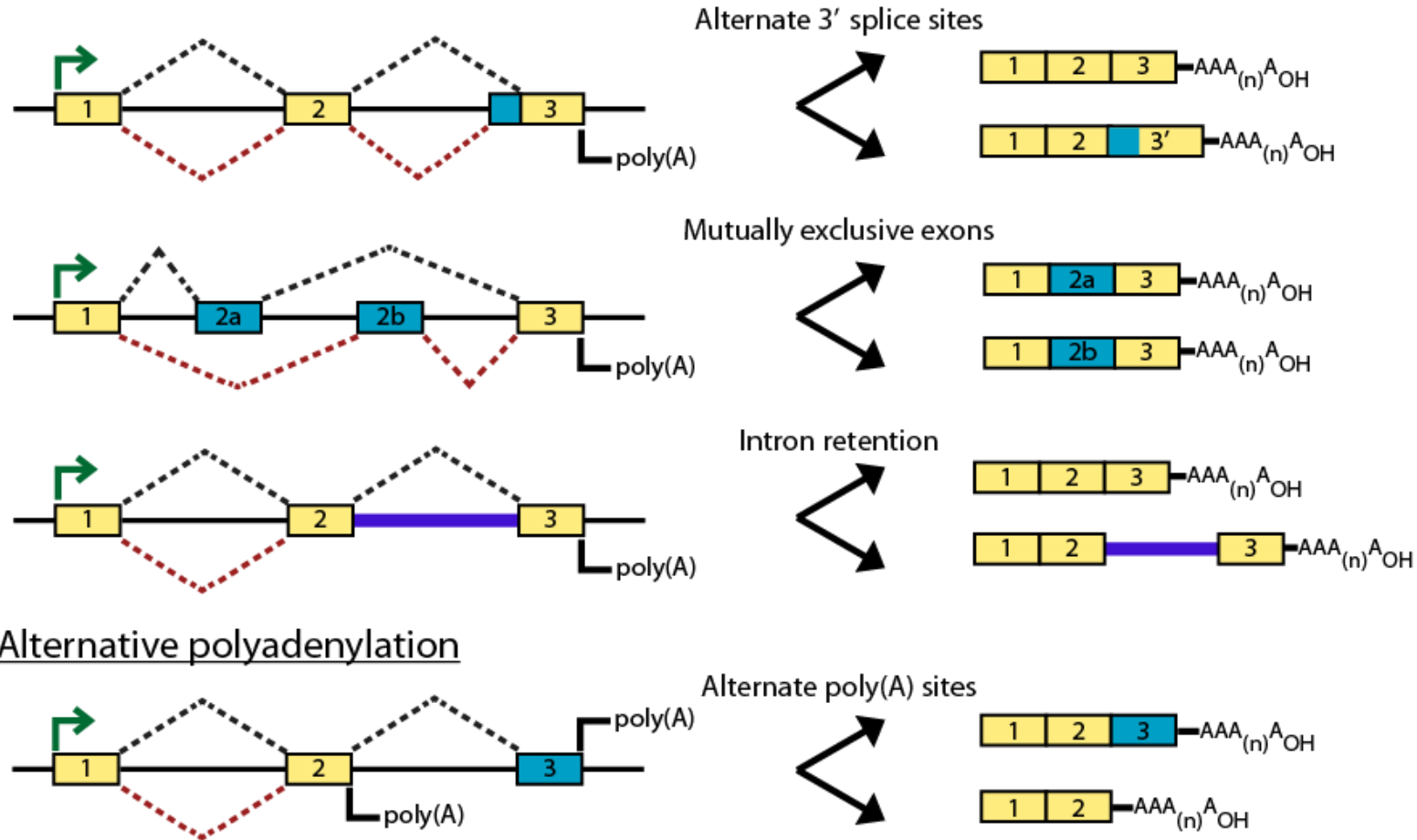


Alternative splicing

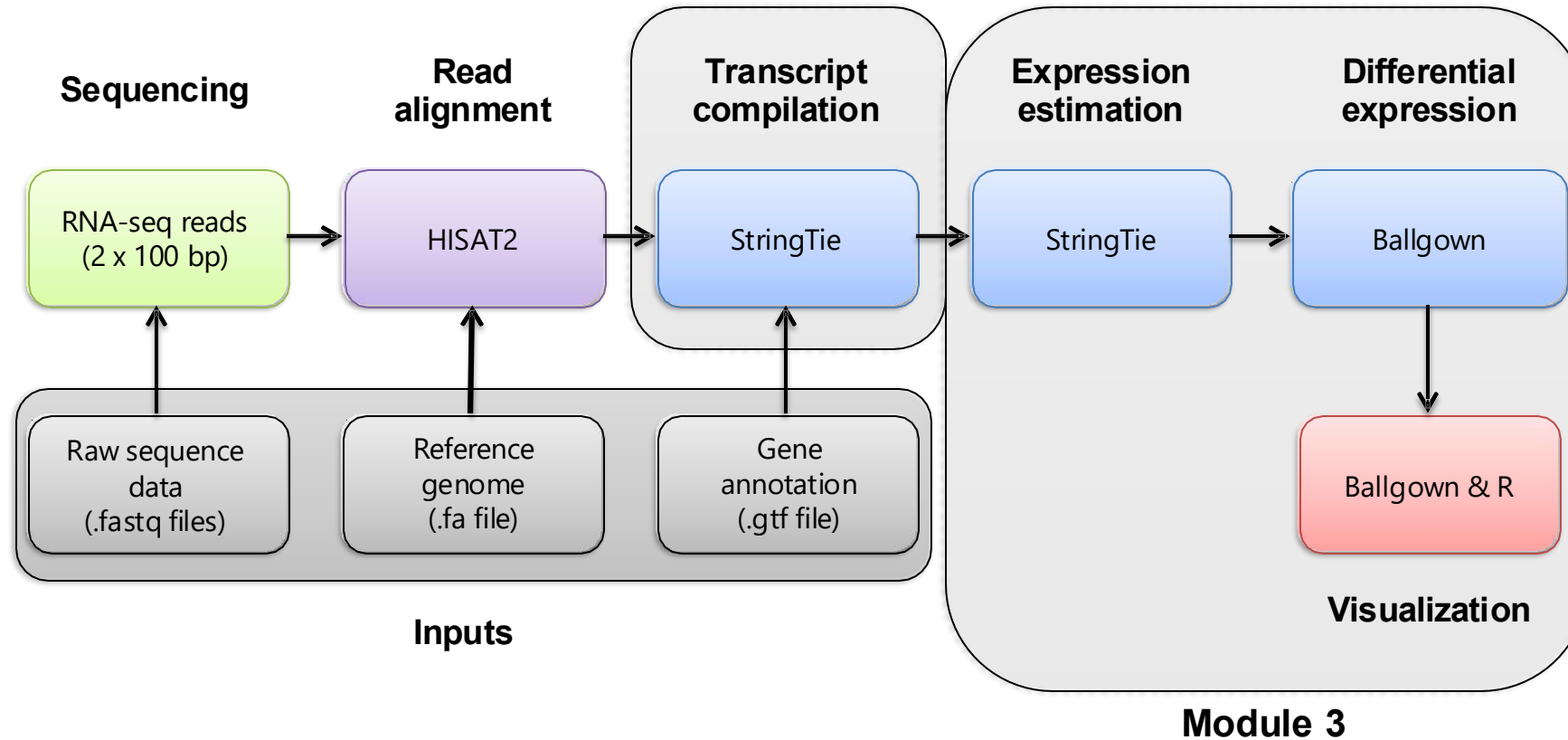


Differential transcript usage

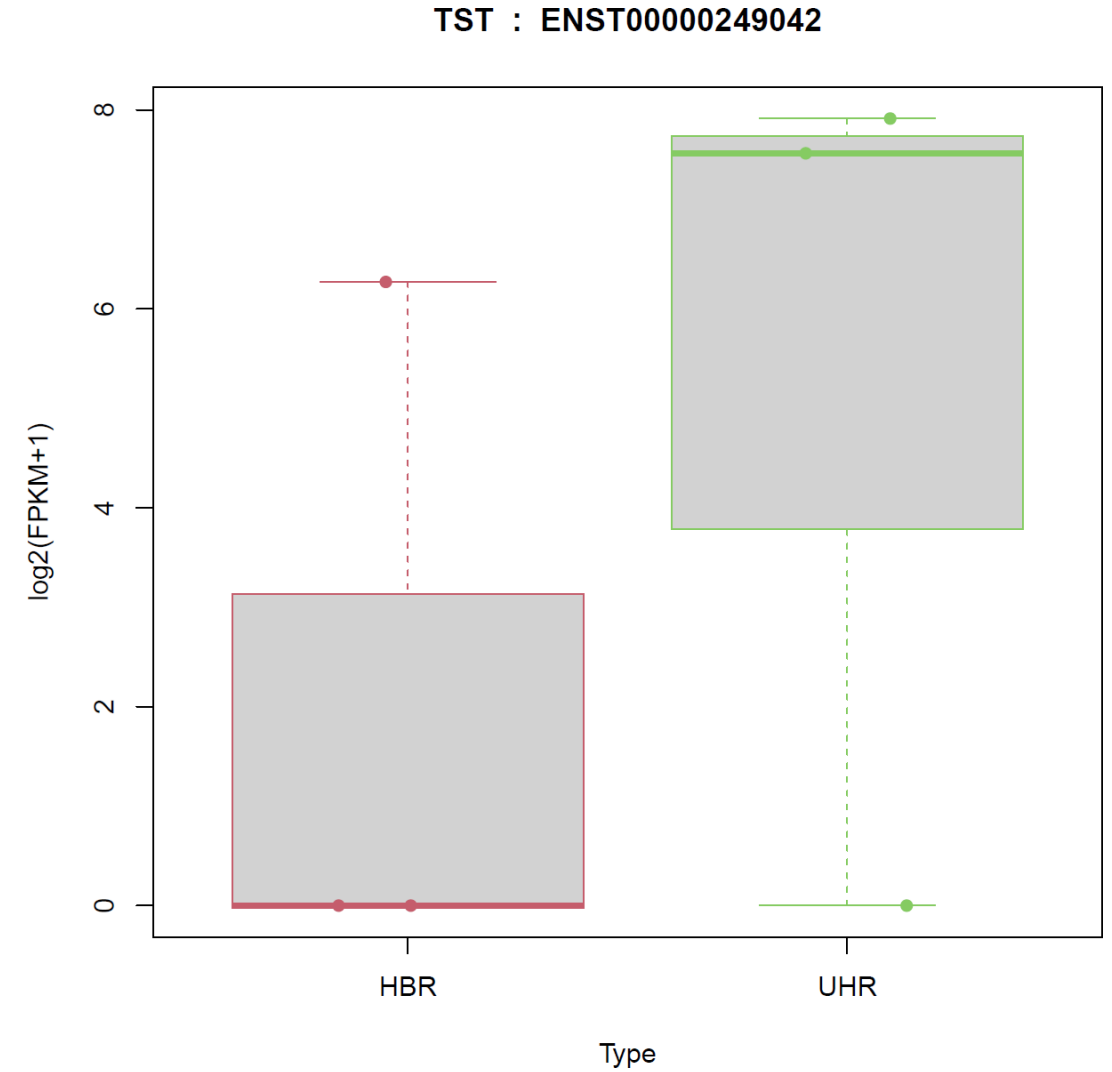
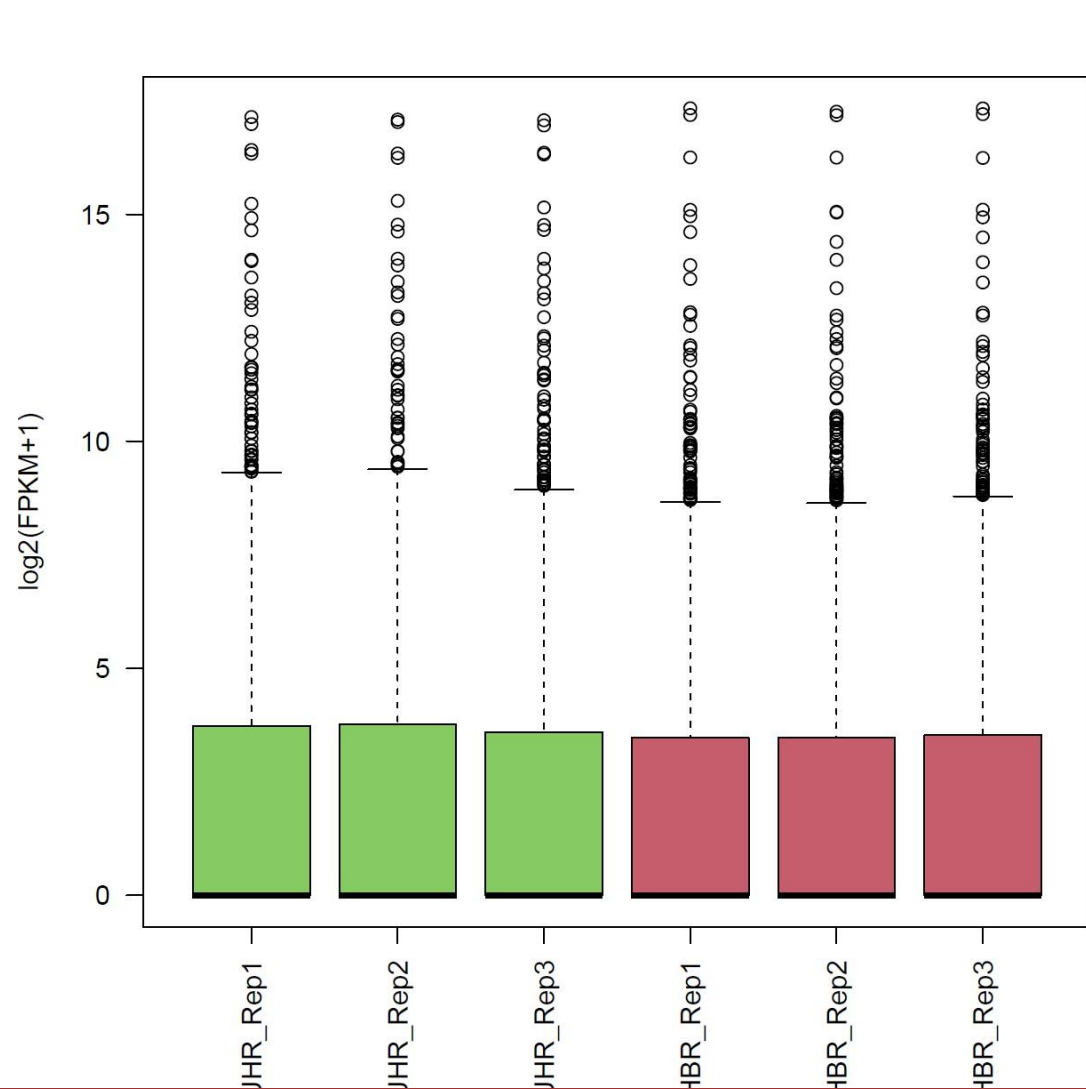
aka alternative splicing



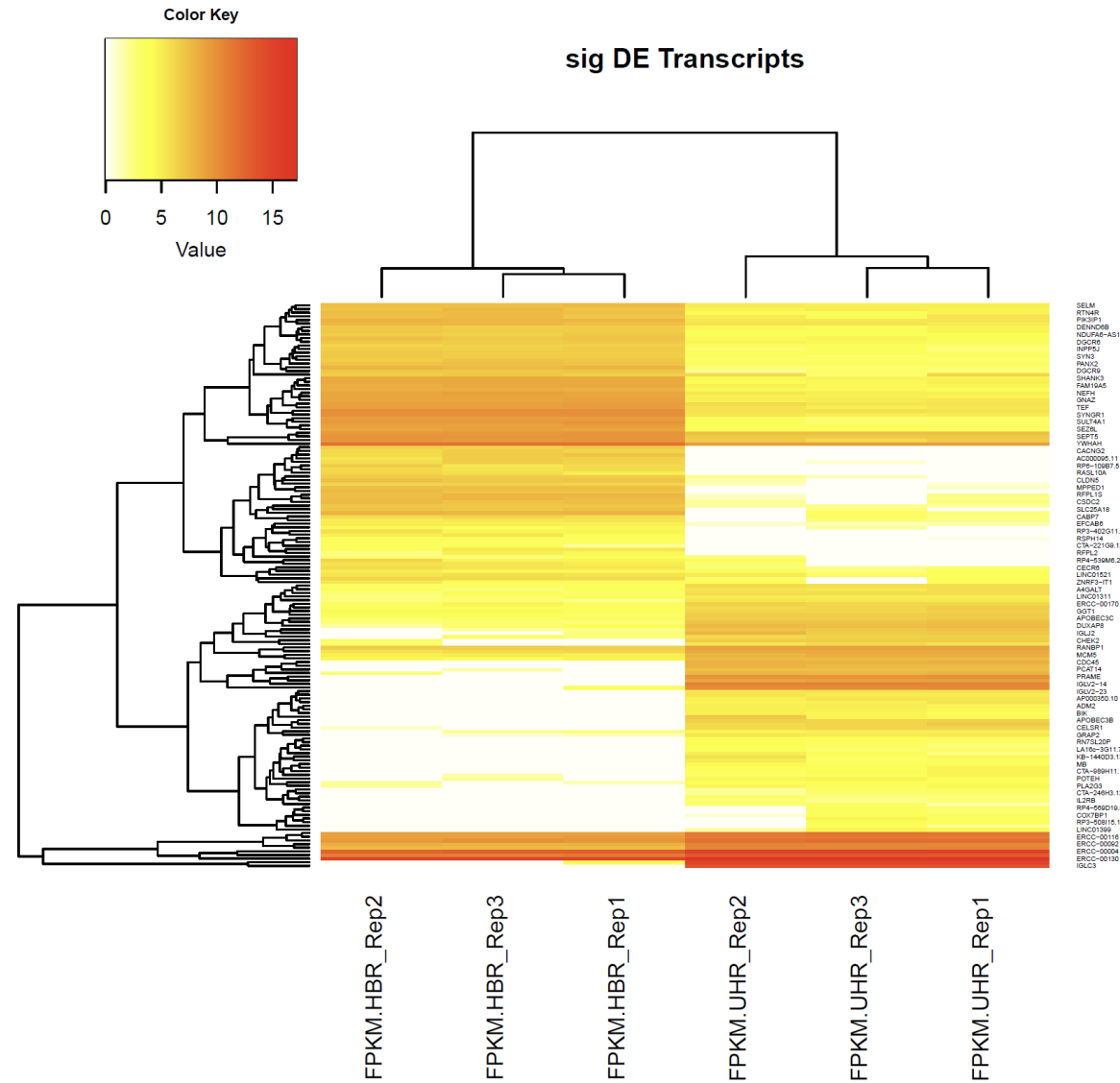
HISAT2/StringTie/Ballgown RNA-seq Pipeline



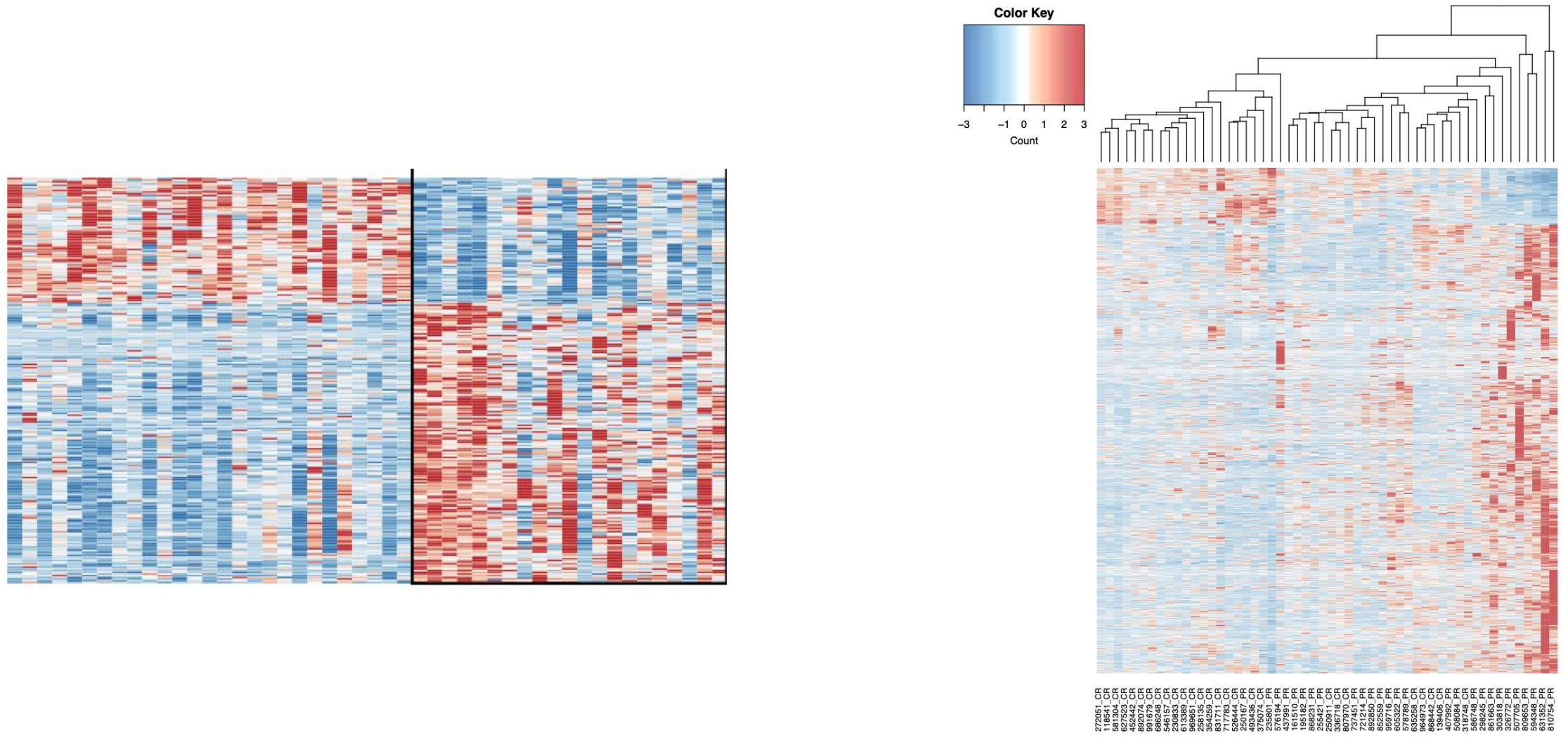
FPKM values for each sample/ across different types



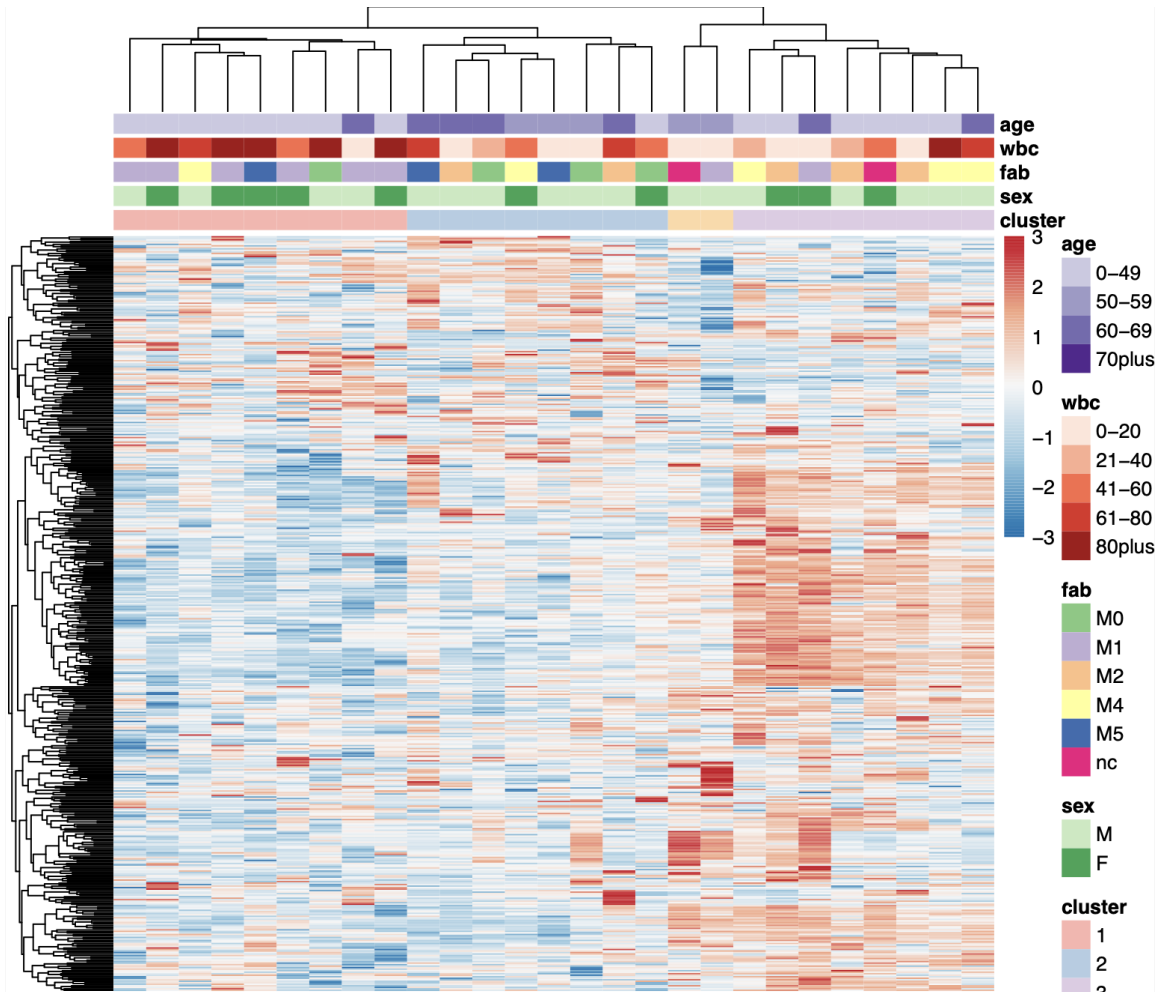
Differential Expression heatmaps



Differential Expression heatmaps



Differential Expression heatmaps



pheatmap is one tool – there are many

DE_Volcano Plot

UHR vs HBR



Downstream interpretation of expression analysis

- Topic for an entire course
- Expression estimates and differential expression lists from StringTie, Ballgown or other alternatives can be fed into many analysis pipelines
- See supplemental R tutorial for how to format expression data and start manipulating in R
- Clustering/Heatmaps
 - Provided by Ballgown
 - For more customized analysis various R packages exist:
 - hclust, heatmap.2, plotrix, ggplot2, etc.
- Classification
 - For RNA-seq data we still rarely have sufficient sample size and clinical details but this is changing
 - Weka is a good learning tool
 - RandomForests R package (biostar tutorial being developed)
- Pathway analysis
 - GSEA, IPA, Cytoscape, many R/BioConductor packages:
<http://www.bioconductor.org/help/search/index.html?q=pathway>

https://genviz.org/module-04-expression/0004/01/01/Expression_Profiling_and_Visualization/

Enrichment analysis



Enrichr

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106,052,179 sets analyzed
449,092 terms
222 libraries

Analyze

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Input data

Expand a gene, a term, or a variant into a gene set:

e.g. STAT3, breast cancer, or rs28897756



Try an example STAT3 breast cancer rs28897756

Include the top 100 most relevant genes



Drop a file or paste a set of Entrez gene symbols on each row in the textbox below. You can try a gene set **example**. Also, you can now try adding a **background**.

Drop a file or paste a set of valid Entrez gene symbols (e.g. STAT3) on each row in the text-box

0 gene(s) entered

pathway, gene ontology, etc

maayanlab.cloud/Enrichr/

Don't forget to input a background (all genes detected by the assay)!

Enrichment analysis



Enrichr

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106,052,204 sets analyzed
449,092 terms
222 libraries

Analyze

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Input data

Expand a gene, a term, or a variant into a gene set:

e.g. STAT3, breast cancer, or rs28897756



Try an example **STAT3** **breast cancer** **rs28897756**

Include the top 100 most relevant genes



Drop a file or paste a set of Entrez gene symbols on each row in the textbox below. You can try a gene set **example**. Also, you can now try adding a **background** (**clear**).

Drop a file or paste a set of valid Entrez gene symbols (e.g. STAT3) on each row in the text-box

A1BG
A2M
NAT1
NAT2
SERPINA3
AADAC
AAMP
AANAT
AARS1
ABAT

0 gene(s) entered

pathway, gene ontology, etc

maayanlab.cloud/Enrichr/

Don't forget to input a background (all genes detected by the assay)!

Enrichment analysis



Enrichr

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Description Sample gene set (614 genes)  

Reactome Pathways 2024



Regulation of Beta-Cell Development
Neuronal System
Digestion and Absorption
Digestion
Regulation of Gene Expression in Beta Cells

WikiPathways 2024 Human



Pancreatic Cancer Subtypes WP5390
6Q16 Copy Number Variation WP5400
PTF1A Related Regulatory Pathway WP4147
Peptide GPCRs WP24
GLP 1 From Intestine And Pancreas Role In t

BioPlanet 2019



Pancreatic secretion
Maturity onset diabetes of the young
Pancreatic beta-cell development regulation
Gene expression regulation in pancreatic be
Neuroactive ligand-receptor interaction

WikiPathways 2024 Mouse



Ptf1a Related Regulatory Pathway WP201
Dravet Syndrome Scn1a A1783V Point Mutat
GPCRs Peptide WP234
GPCRs Non Odorant WP1396
SIDS Susceptibility Pathways WP1266

KEGG 2021 Human



Pancreatic secretion
Maturity onset diabetes of the young
Neuroactive ligand-receptor interaction
Protein digestion and absorption
Carbohydrate digestion and absorption

ARCHS4 Kinases Coexp



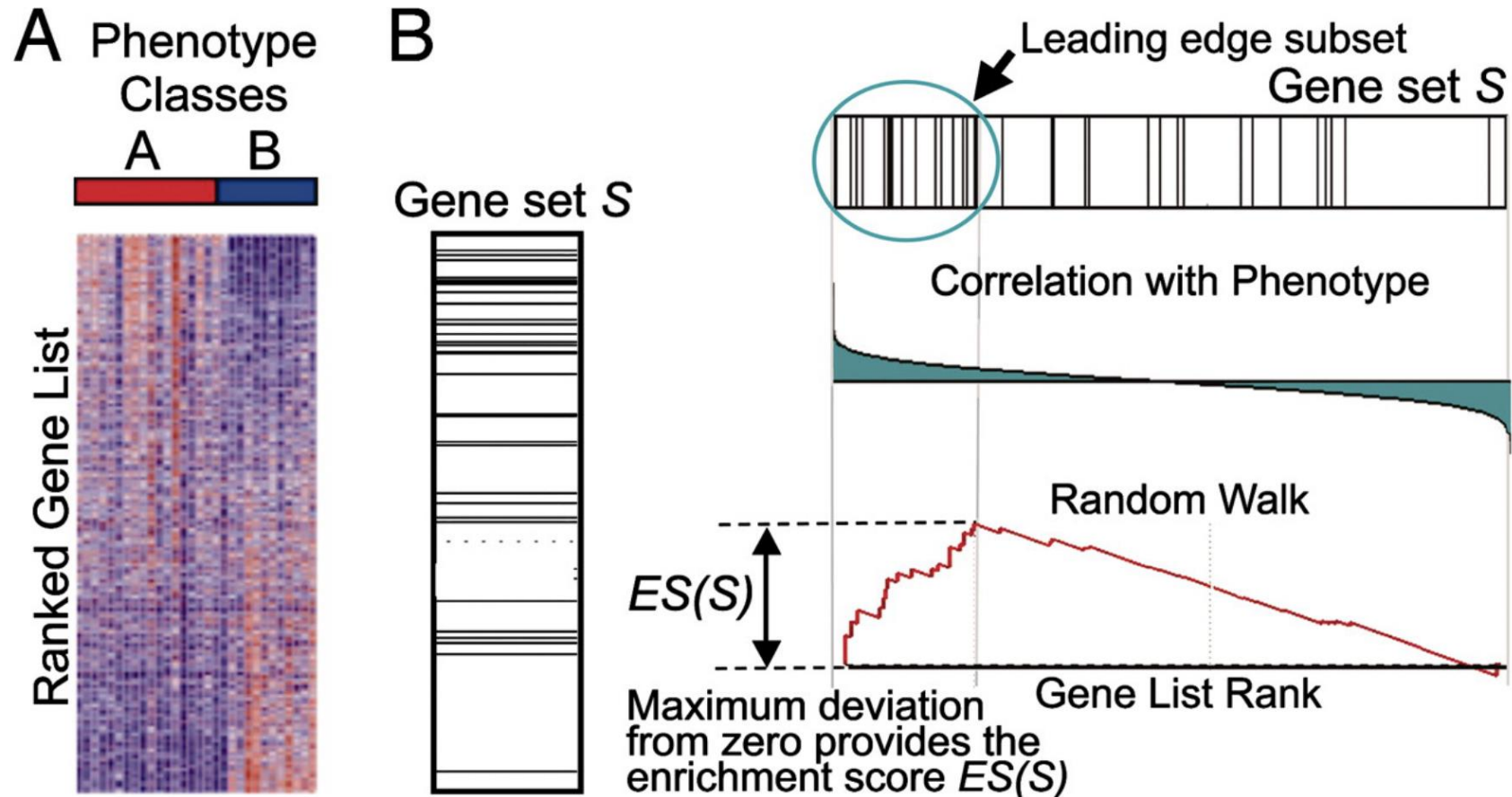
ACVR1C human kinase ARCHS4 coexpression
MYO3A human kinase ARCHS4 coexpression
SGK2 human kinase ARCHS4 coexpression
PDK4 human kinase ARCHS4 coexpression
ERBB3 human kinase ARCHS4 coexpression

pathway, gene ontology, etc

maayanlab.cloud/Enrichr/

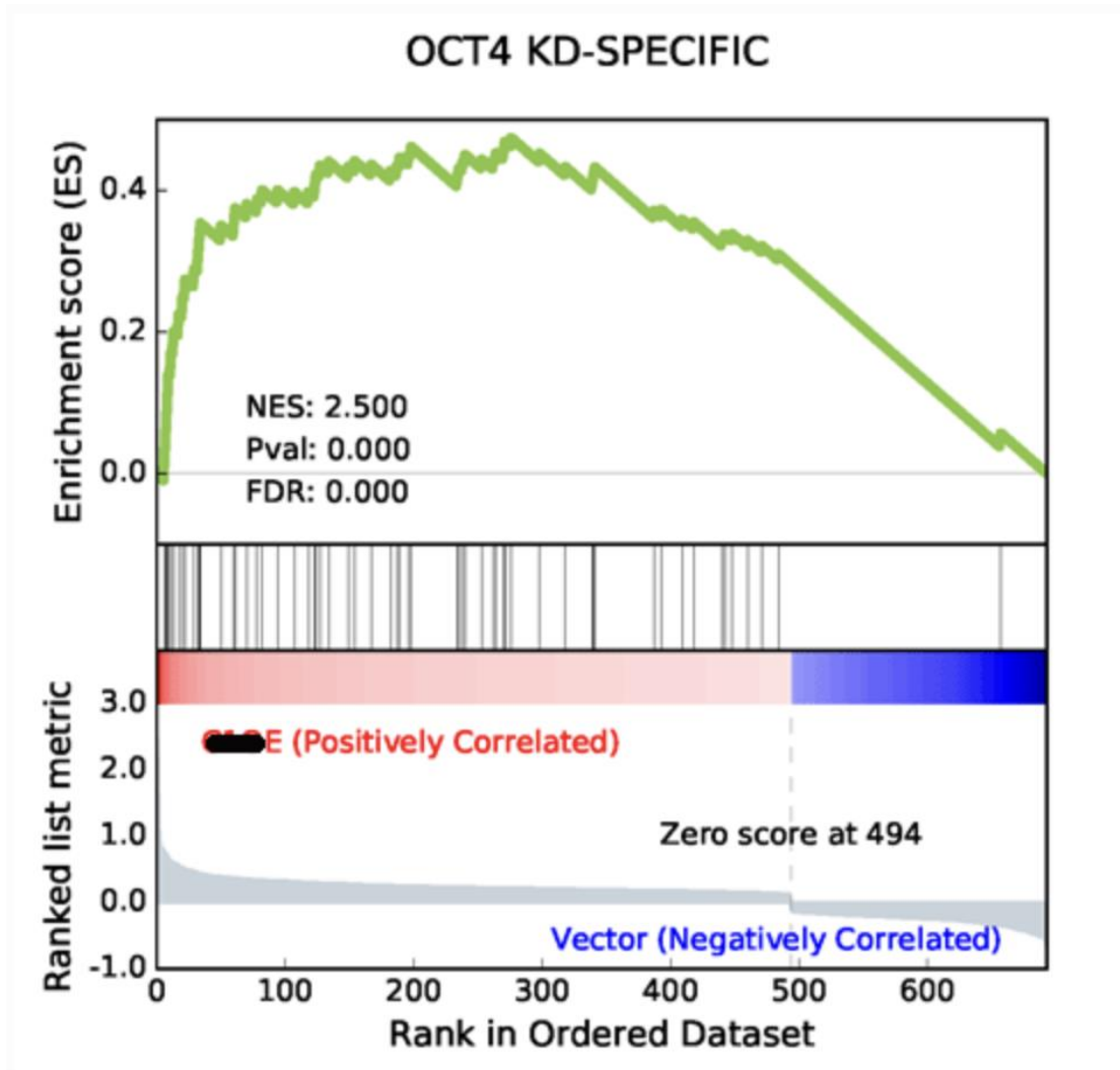
Don't forget to input a background (all genes detected by the assay)!

GSEA – gene set enrichment analysis



Subramanian et al., PNAS 102(43), 15545–15550 (2005).

GSEA – gene set enrichment analysis



GSEA – gene set enrichment analysis



Human Collections

H

hallmark gene sets are coherently expressed signatures derived by aggregating many MSigDB gene sets to represent well-defined biological states or processes.

C5

ontology gene sets consist of genes annotated by the same ontology term.

C1

positional gene sets corresponding to human chromosome cytogenetic bands.

C6

oncogenic signature gene sets defined directly from microarray gene expression data from cancer gene perturbations.

C2

curated gene sets from online pathway databases, publications in PubMed, and knowledge of domain experts.

C7

immunologic signature gene sets represent cell states and perturbations within the immune system.

C3

regulatory target gene sets based on gene target predictions for microRNA seed sequences and predicted transcription factor binding sites.

C8

cell type signature gene sets curated from cluster markers identified in single-cell sequencing studies of human tissue.

C4

computational gene sets defined by mining large collections of cancer-oriented expression data.

GSEA – gene set enrichment analysis

HALLMARK_ADIPOGENESIS	HALLMARK_G2M_CHECKPOINT	HALLMARK_NOTCH_SIGNALING
HALLMARK_ALLOGRAFT_REJECTION	HALLMARK_GLYCOLYSIS	HALLMARK_OXIDATIVE_PHOSPHORYLATION
HALLMARK_ANDROGEN_RESPONSE	HALLMARK_HEDGEHOG_SIGNALING	HALLMARK_P53_PATHWAY
HALLMARK_ANGIOGENESIS	HALLMARK_HEME_METABOLISM	HALLMARK_PANCREAS_BETA_CELLS
HALLMARK_APICAL_JUNCTION	HALLMARK_HYPOXIA	HALLMARK_PEROXISOME
HALLMARK_APICAL_SURFACE	HALLMARK_IL2_STAT5_SIGNALING	HALLMARK_PI3K_AKT_MTOR_SIGNALING
HALLMARK_APOPTOSIS	HALLMARK_IL6_JAK_STAT3_SIGNALING	HALLMARK_PROTEIN_SECRETION
HALLMARK_BILE_ACID_METABOLISM	HALLMARK_INFLAMMATORY_RESPONSE	HALLMARK_REACTIVE_OXYGEN_SPECIES_PATHWAY
HALLMARK_CHOLESTEROL_HOMEOSTASIS	HALLMARK_INTERFERON_ALPHA_RESPONSE	HALLMARK_SPERMATOGENESIS
HALLMARK_COAGULATION	HALLMARK_INTERFERON_GAMMA_RESPONSE	HALLMARK_TGF_BETA_SIGNALING
HALLMARK_COMPLEMENT	HALLMARK_KRAS_SIGNALING_DN	HALLMARK_TNFA_SIGNALING_VIA_NFKB
HALLMARK_DNA_REPAIR	HALLMARK_KRAS_SIGNALING_UP	HALLMARK_UNFOLDED_PROTEIN_RESPONSE
HALLMARK_E2F_TARGETS	HALLMARK_MITOTIC_SPINDLE	HALLMARK_UV_RESPONSE_DN
HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION	HALLMARK_MTORC1_SIGNALING	HALLMARK_UV_RESPONSE_UP
HALLMARK_ESTROGEN_RESPONSE_EARLY	HALLMARK_MYC_TARGETS_V1	HALLMARK_WNT_BETA_CATENIN_SIGNALING
HALLMARK_ESTROGEN_RESPONSE_LATE	HALLMARK_MYC_TARGETS_V2	HALLMARK_XENOBIOTIC_METABOLISM
HALLMARK_FATTY_ACID_METABOLISM	HALLMARK_MYOGENESIS	

<https://www.gsea-msigdb.org/gsea/msigdb>

GSEA – gene set enrichment analysis

Human Gene Set: HALLMARK_KRAS_SIGNALING_DN

For the Mouse gene set with the same name, see [HALLMARK_KRAS_SIGNALING_DN](#)

Standard name	HALLMARK_KRAS_SIGNALING_DN
Systematic name	M5956
Brief description	Genes down-regulated by KRAS activation.
Full description or abstract	
Collection	H: Hallmark
Source publication	Pubmed 26771021 Authors: Liberzon A,Birger C,Thorvaldsdóttir H,Ghandi M,Mesirov JP,Tamayo P
Exact source	
Related gene sets	(show 49 additional gene sets from the source publication) (show 74 gene sets from any of these authors) (show 16 founder gene sets for this hallmark gene set)
External links	
Filtered by similarity ?	
Source species	Homo sapiens
Contributed by	Arthur Liberzon (MSigDB Team)
Source platform or identifier namespace	HUMAN_GENE_SYMBOL
Dataset references	(show 5 hallmark refinement datasets) (show 1 hallmark validation datasets)
Download gene set	format: grp gmt xml json TSV metadata
Compute overlaps ?	(show collections to investigate for overlap with this gene set)

<https://www.gsea-msigdb.org/gsea/msigdb>

Assignment